9 **(a)** A resistance wire of uniform cross-sectional area 3.3×10^{-7} m^2 and length 2.0 m is made of metal of resistivity 5.0×10^{-7} Ω m.

(i) Determine the resistance of the wire.

resistance = ……..Ω [1]

(ii) The ends of the resistance wire are connected to the terminals S and T in the circuit shown in Figure 9.1.

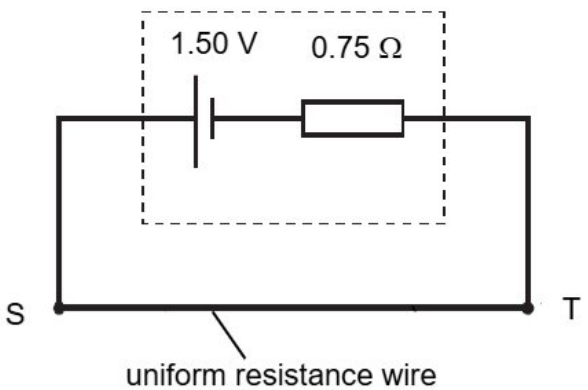


Figure 9.1

The cell has an e.m.f. of 1.50 V and internal resistance 0.75 Ω.

The number density of charge carriers in the wire is 8.5×10^{28} m^{-3}.

Calculate the average drift speed of the charge carriers.

speed = ……..m s^{-1} [2]

(iii) The wire in **9(a)(ii)** is now replaced with another wire of the same material and diameter but twice the length. The temperature of both wires is the same.

State and explain the change, if any, to the drift velocity of the charge carriers in the second wire.

.....

.....

.....

.....

.....[3]

(b) The variation with potential difference V of current I for a light emitting diode (LED) is shown in Figure 9.2.

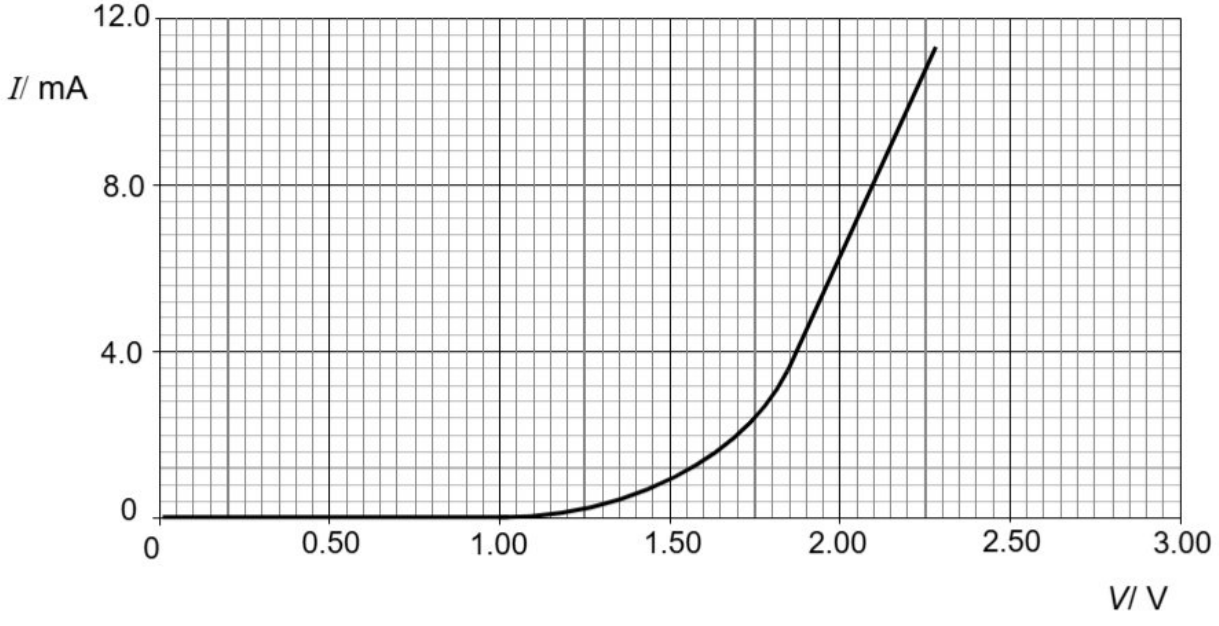


Figure 9.2

(i) Use Figure 9.2 to determine the resistance of the LED at 2.25 V.

resistance = ...Ω [2]

(ii) On Figure 9.2, draw the $I - V$ characteristics of a 160 Ω ohmic resistor. Label the line R.

[1]

(iii) Two of these LEDs are connected to a 3.0 V battery with negligible internal resistance and an ohmic 160 Ω resistor as shown in Figure 9.3.

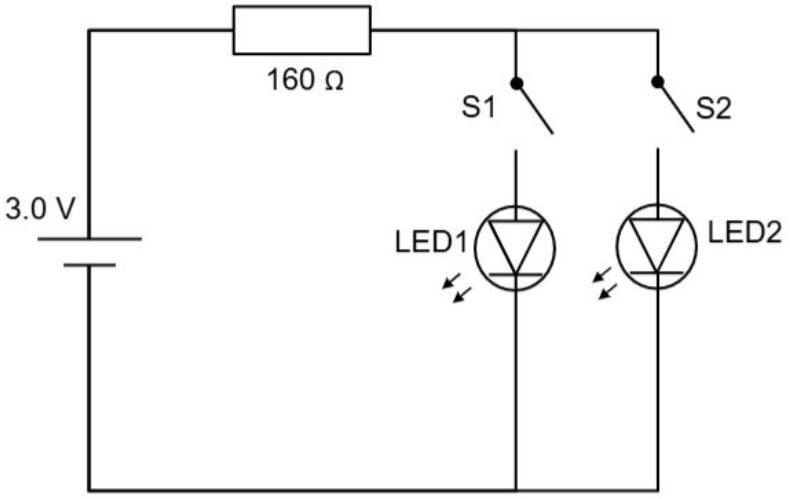


Figure 9.3

1. Determine the current in LED1 when switch S1 is closed and switch S2 remains open. Show your working clearly.

current = ...A [2]

2. Explain why LED1 becomes dimmer when S2 is also closed.

.....

.....

.....

.....

.....[3]

(c) A sinusoidal alternating voltage supply, with a peak value of 150 V, is applied across the primary coil of a transformer. There are 1200 turns in the primary coil. The variation with time t of the e.m.f E induced across the secondary coil is shown in Figure 9.4.

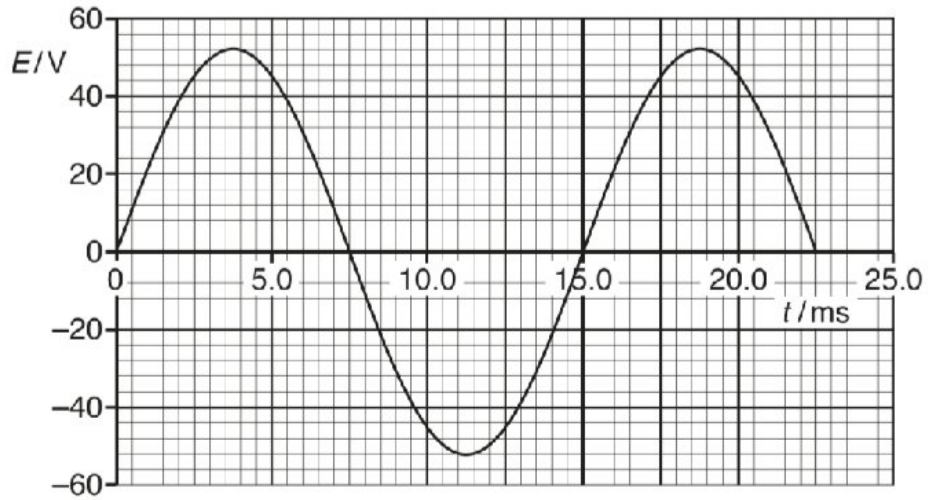


Figure 9.4

(i) Use data from Figure 9.4 to calculate the number of turns in the secondary coil.

number of turns = ……..…….. [2]

(ii) A resistor is connected between the output terminals of the secondary coil. The mean power dissipated in the resistor is 1.2 W. It may be assumed that the varying voltage across the resistor is equal to the varying e.m.f. E shown in Figure 9.4.

1. Calculate the resistance of the resistor.

resistance = ……Ω [2]

2. On Figure 9.5, sketch the variation with time t of the power P dissipated in the resistor for $t = 0$ to $t = 22.5$ ms.

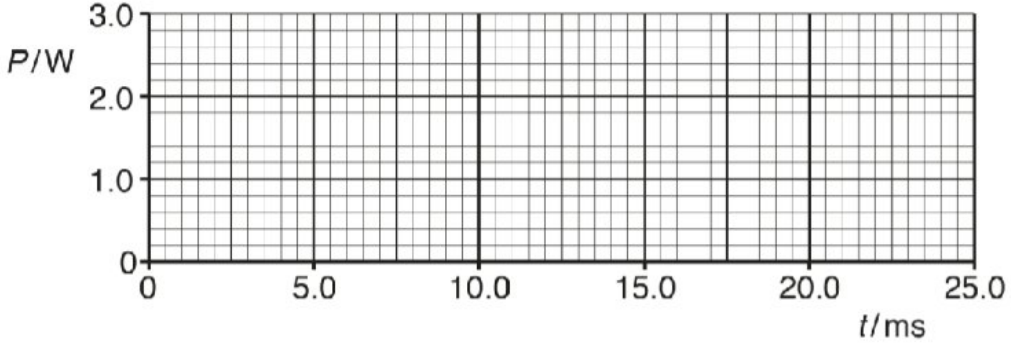


Figure 9.5

[2]

[Total: 20]